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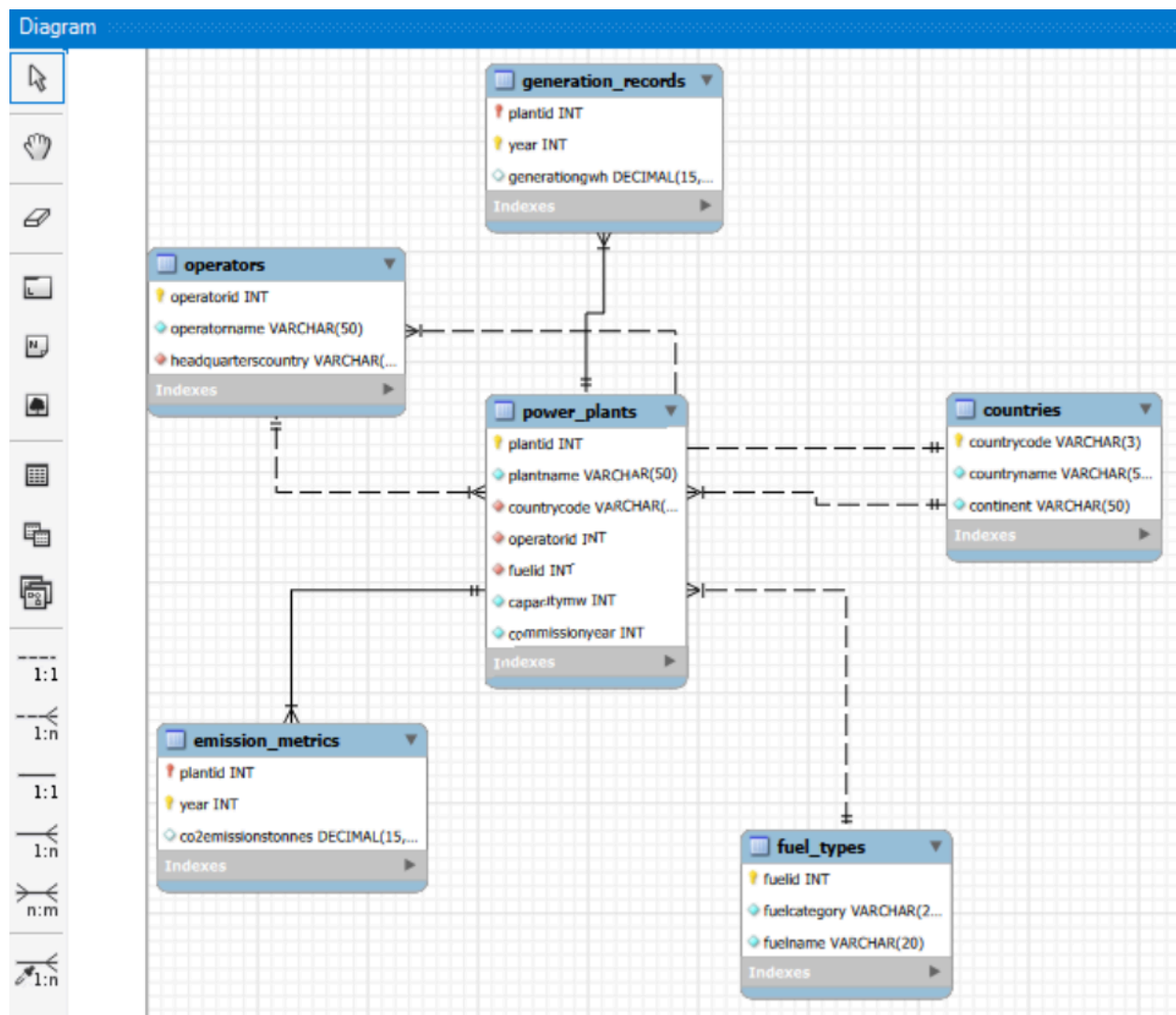
CSC 351

5 October 2026

Homework Assignment 3

Building a Global Energy and Power Plants Relational Model in MySQL

Full Diagram:



SQL Queries & Screenshots

SQL Queries:

1. Retrieve the plant name, country name, operator name, fuel category, fuel name, capacity (in MW), and commission year for all power plants without using table aliases. Sort the results in descending order by capacity.

a. `select`

```
power_plants.plantname,  
countries.countryname,  
operators.operatorname,  
fuel_types.fuelcategory,  
fuel_types.fuelname,  
power_plants.capacitymw,  
power_plants.commissionyear
```

`from power_plants`

`inner join countries on power_plants.countrycode = countries.countrycode`

`inner join operators on power_plants.operatorid = operators.operatorid`

`inner join fuel_types on power_plants.fuelid = fuel_types.fuelid`

`order by power_plants.capacitymw desc;`

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
▶	Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
	Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
	Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
	Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
	Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
	Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
	Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
	Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

2. Retrieve the plant name, country code, calendar year, and annual generation (in GWh) for all power plants for the year 2024 without using table aliases. Sort the results in descending order by generation.

a. `select`

```
power_plants.plantname,  
power_plants.countrycode,  
generation_records.year,  
generation_records.generationgwh
```

`from power_plants`

`innerjoin generation_records on power_plants.plantid`

`generation_records.plantid`

`where generation_records.year = 2024`

order by generation_records.generationgwh desc;

Result Grid   Filter Rows: Export: 				
	plantname	countrycode	year	generationgwh
▶	Three Gorges	CHN	2024	98800.00
	Itaipu	BRA	2024	89000.10
	Tuocketuo	CHN	2024	71000.30
	Vindhyachal	IND	2024	35600.80
	Grand Coulee	USA	2024	21500.50
	Janschwalde	DEU	2024	18400.20
	Loy Yang	AUS	2024	14200.40
	Kashiwazaki-Kariwa	JPN	2024	0.00

3. Retrieve the plant name, country code, calendar year, annual generation (in GWh), and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table aliases. Sort the results in ascending order by CO2 emissions.

a. select

```
power_plants.plantname,  
power_plants.countrycode,  
generation_records.year,  
generation_records.generationgwh,  
emission_metrics.co2emissionstonnes
```

```
from power_plants
```

```
inner join generation_records on power_plants.plantid =  
generation_records.plantid
```

```
inner join emission_metrics on power_plants.plantid = emission_metrics.plantid  
and generation_records.year = emission_metrics.year
```

```
where generation_records.year = 2024
```

```
order by emission_metrics.co2emissionstonnes;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: ☐					
	plantname	countrycode	year	generationgwh	co2emissionstonnes
▶	Grand Coulee	USA	2024	21500.50	0.00
	Itaipu	BRA	2024	89000.10	0.00
	Kashiwazaki-Kariwa	JPN	2024	0.00	0.00
	Three Gorges	CHN	2024	98800.00	0.00
	Loy Yang	AUS	2024	14200.40	1950000.00
	Janschwalde	DEU	2024	18400.20	2450000.00
	Vindhyachal	IND	2024	35600.80	3100000.00
	Tuocketuo	CHN	2024	71000.30	5200000.00

4. Write a SQL query using a **Common Table Expression (CTE)** to calculate the total cumulative power generation (in GWh) for each operator across all available years without using table aliases. Retrieve the operator name, headquarters country, and their total generated power, and sort the results in descending order by total generation.

a. `with operator_generation_cte as (
 select
 operators.operatorname,
 operators.headquarterscountry,
 generation_records.generationgwh
 from operators
 inner join power_plants on operators.operatorid = power_plants.operatorid
 inner join generation_records on power_plants.plantid
 generation_records.plantid
)`
`select
 operator_generation_cte.operatorname,
 operator_generation_cte.headquarterscountry,
 sum(operator_generation_cte.generationgwh) as total_generation
from operator_generation_cte
group by operator_generation_cte.operatorname,
operator_generation_cte.headquarterscountry
order by total_generation;`

Result Grid  Filter Rows: Export:  Wra			
	operatorname	headquarterscountry	total_generation
▶	China Yangtze Power	CHN	169800.30
	Electrobras	BRA	89000.10
	NTPC Limited	IND	35600.80
	Global Hydro Inc	USA	21500.50
	Rhein Power AG	DEU	18400.20
	Snowy Hydro	AUS	14200.40
	Tepco Energy	JPN	0.00

5. Write a SQL query using **two separate Common Table Expressions (CTEs)**--one to calculate total power generation by country code and another to calculate total CO2 emissions by country code without using table aliases. Then, join these CTEs with the countries table to display the country name, total generation (in GWh), and total CO2 emissions (in tonnes). Sort the results in descending order by total generation.

a. `with country_generation_cte as (
 select
 power_plants.countrycode,
 generation_records.generationgwh
 from power_plants
 inner join generation_records on power_plants.plantid =
 generation_records.plantid
),
country_emissions_cte as (
 select
 power_plants.countrycode,
 emission_metrics.co2emissionstonnes
 from power_plants
 inner join emission_metrics on power_plants.plantid =
 emission_metrics.plantid
)
select
 countries.countryname,
 sum(country_generation_cte.generationgwh) as total_generation,
 sum(country_emissions_cte.co2emissionstonnes) as total_emissions
from countries`

```

inner join country_generation_cte on countries.countrycode =
country_generation_cte.countrycode
inner join country_emissions_cte on countries.countrycode
country_emissions_cte.countrycode
group by countries.countryname
order by total_generation desc;

```

Result Grid  Filter Rows: Export			
	countryname	total_generation	total_emissions
▶	China	339600.60	10400000.00
	Brazil	89000.10	0.00
	India	35600.80	3100000.00
	United States	21500.50	0.00
	Germany	18400.20	2450000.00
	Australia	14200.40	1950000.00
	Japan	0.00	0.00